THE RPM OF CHANGE

THE TRANSFER OF TECHNOLOGY BY THE US FOR THE MANUFACTURE OF THE F414 JET ENGINE WILL ENABLE INDIA TO ABSORB IMMENSE TECHNICAL KNOWHOW TO PRODUCE HI-TECH ENGINES OF ITS OWN IN FUTURE



It is a vision to be hoped for—a sixth generation indigenous fighter jet of the Indian Air Force (IAF) screaming through the sky, its contrail a jagged white against the blue, and an India-made engine responding gamely to the demands of high speed propulsion as well as power and cooling for the aircraft's advanced sensors and weapons systems. To make the above scene a reality by 2047, Indian aeronautical engineers and designers have their task cut out. The most crucial part of a jet aircraft—its engine—has proven to be a hurdle for most. Consider this fact: there are around 40 aircraft manufacturers in the world, but only four countries—the US, UK, Russia and France—have the technology to completely design, develop and produce military aircraft engines. The inability of the indigenous Kaveri engine, developed by the Defence Research and Development Organisation, to meet the parameters to power a fighter jet delayed India's Light Combat Aircraft (LCA) fighter jet programme. LCA Tejas manufacturers Hindustan Aeronautics Ltd (HAL) had to settle for F404 engines made by the US firm GE Aerospace. Recently, the US has agreed to transfer 80 per cent jet engine technology to India for the joint production of the more powerful GE F414 engine. The GE F414 engines will power HAL's LCA Tejas Mk-2 fighter jet.

WHY IT IS A GAME CHANGER

The India specific GE F414 INS6 engines will enable the LCA Mk2 to have advanced sensors, a large area display cockpit, better operational capability with increased payload, and reduced turnaround time for repair and maintenance. It will also prove to be a game-changer because the LCA Mark-2 jet will replace the IAF's 16 fighter jet squadrons, including three squadrons of Mirage 2000s, five squadrons of MiG-29s, six Jaguar squadrons and the two remaining MiG-21 Bison squadrons. The jet engine deal, announced during PM Narendra Modi's state visit to the US in June, enables the IAF to look at a combat strength of about 40 fighter squadrons by 2040. Significantly, it affords India self-reliance in critical technologies and reduces its dependence on imports.

To begin with, 99 GE F414 engines will be manufactured in India. The IAF has expressed a need for around 600 India-made fighter jets for its fleet, so HAL would be ultimately required to manufacture a requisite number of engines. There is also a pos-

CHALLENGES

- To attain full technological 'knowhow' for manufacturing jet engines
- India still lacks the manufacturing/technological depth to fully indigenise a jet engine
- Dependence on foreign sources to continue until a complete design for a jet engine is developed